PIP104 University Project-II
VIVA-VOCE

PEDIATRIC TELE DERMATOLOGY

Batch Number:CSE-G02

Roll Number	Student Name
20211CSE0823	MD Haseebuddin
20211CSE0808	Shaik Ahmed
20211CSE0559	Mohammed Noaman Ahmed

Under the Supervision of,

Ms. Akkamahadevi C
Assistant Professor
School of Computer Science & Engineering
Presidency University

Introduction

- The diagnosis of neonatal and pediatric skin conditions presents unique challenges, as the patients are highly sensitive and often unable to articulate their symptoms. Skin issues in infants and young children, such as eczema, rashes, birthmarks, or infections, typically exhibit distinct visual characteristics that can be assessed remotely. Given the availability of smartphones and internet access, telemedicine has become a promising solution in this domain.
- Remote dermatological consultations, particularly for pediatric and neonatal cases, leverage image-based assessments. These allow healthcare providers to diagnose and recommend treatment plans without the need for in-person visits. This is particularly valuable in regions with limited access to specialized care or during situations where face-to-face consultations may not be feasible. By enabling parents or caregivers to capture images of the child's skin condition and send them to physicians, healthcare services can be provided quickly and efficiently.
- An application that facilitates the remote exchange of medical images, diagnosis tagging, and response management offers significant advantages for both physicians and patients. Furthermore, incorporating the ability to store images for research purposes could contribute to advances in pediatric dermatology by providing valuable data for future studies. In this context, the focus is not on automating diagnosis but on streamlining the communication process between patients and healthcare providers for more efficient and effective care.

Literature Review

S.No	Name	Date	Description
1	Olivia S. Jew MD, MBA, Aditi S. Murthy MD, Kristen Dunley MD	31st August 2020	**Implementation of a pediatric provider-to-provider store-and-forward teledermatology system** https://onlinelibrary.wiley.com/doi/abs/10.1111/pde.14226
2	Vartan Pahalyants MD, MBA, William S. Murphy MD, MBA, Nicole S. Gunasekera MD, MBA, Shinjita Das MD, Elena B. Hawryluk MD, PhD, Daniela Kroshinsky MD	31st August 2021	**Evaluation of electronic consults for outpatient pediatric patients with dermatologic complaints** https://onlinelibrary.wiley.com/doi/10.1111/pde.14719
3	Jessica L. Crockett, Kelly M. Cordoro	17th October 2024	Pediatric Dermatology eConsultation in Academic Centers: A Collaborative Model for Optimized Outpatient Care https://onlinelibrary.wiley.com/doi/abs/10.1111/pde.15793
4	Valencia Long, MBBS, MRCP, MMed, Nisha Suyien Chandran, MBBS, MRCP, MMed	17th May 2022	A Glance at the Practice of Pediatric Teledermatology Pre- and Post-COVID-19: Narrative Review https://pubmed.ncbi.nlm.nih.gov/35610984/

Research Gaps Identified

- Limited Focus on Pediatric-Specific Needs
- Lack of Comprehensive Diagnostic Tools
- Data Scarcity for Pediatric Cases
- Inadequate Integration of Parent-Child Interaction
- Limited Accessibility in Remote Areas
- Lack of Standardization in Diagnostic Practices
- Privacy and Ethical Concerns
- Limited Longitudinal Studies
- Usability and Engagement Challenges

Proposed Methodology

1. **Mobile App (React Native)**:
    1. Image Capture: Users upload or take photos of skin conditions.
    2. Data Submission: App sends image and patient data to the backend via REST API.
    3. Receive Diagnosis: Users receive diagnosis and feedback from physicians.
2. **Backend (Node.js + Express)**:
    1. REST API: Handles image uploads, stores metadata, and returns diagnosis.
    2. Business Logic: Processes requests and interacts with the database and cloud storage.
    3. Security: Ensures secure communication (JWT, HTTPS).
3. **MySQL Database**:
    1. Stores patient information, image metadata, and physician diagnosis tags.
    2. Facilitates data retrieval for research purposes.
4. **Azure Blob Storage**:
    1. Securely stores uploaded images.
    2. Provides scalable cloud storage with access control.

Objectives

1. Develop an Image-Based Diagnosis Platform

2. Enable Physician-Tagged Diagnosis

3. Automate Image Storage for Research

4. Provide Secure Communication for Feedback

5. Enhance Usability for Caregivers

6. Ensure Regulatory Compliance and Data Privacy

System Design & Implementation

1. **Requirements Gathering**

2. System Design

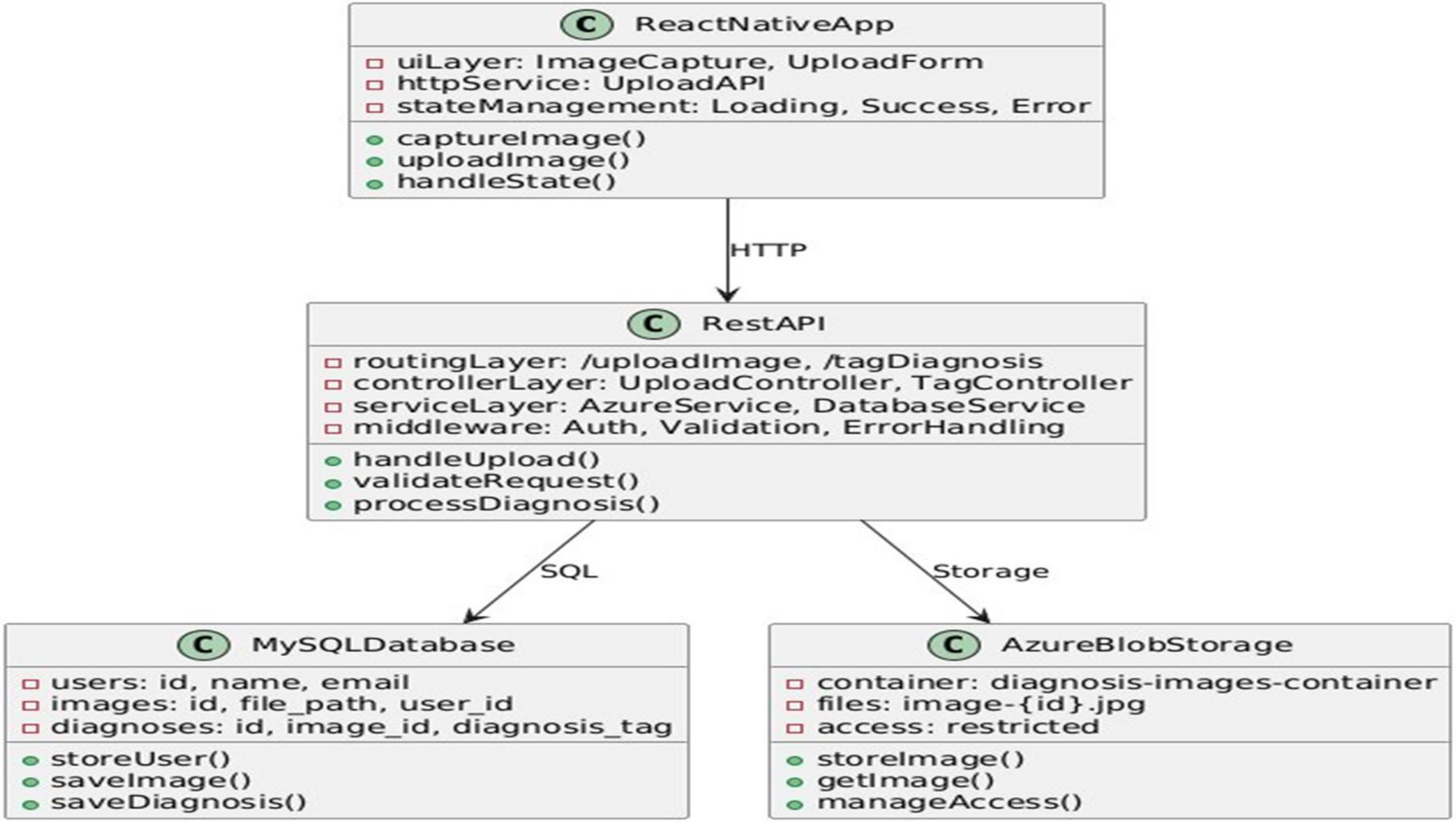
- Architecture Design
- Database Design
- User Interface Design

3. App Development

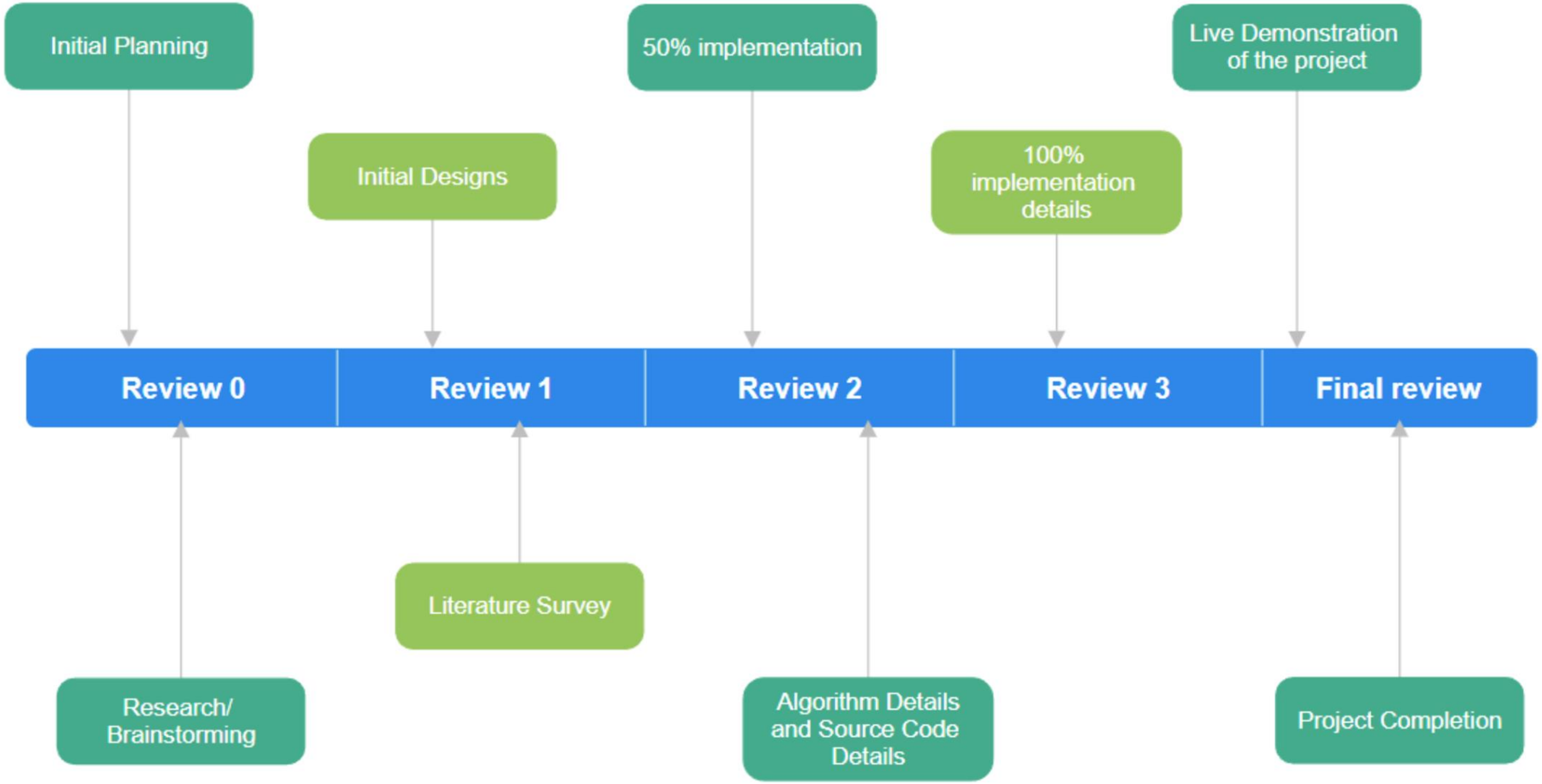
- Frontend Development
- Backend Development

- This structured methodology ensures the system is developed to meet the requirements of caregivers and physicians while maintaining high security, usability, and scalability.

Architecture



Timeline of Project



Outcomes

1. Remote Diagnosis:

- Efficient diagnosis and feedback via the app, without face-to-face interaction.

2. Secure Storage:

- Images stored securely in Azure, with metadata in MySQL for research use.

3. Scalable and Accessible:

- Scalable cloud-based system, accessible for patients in remote areas.

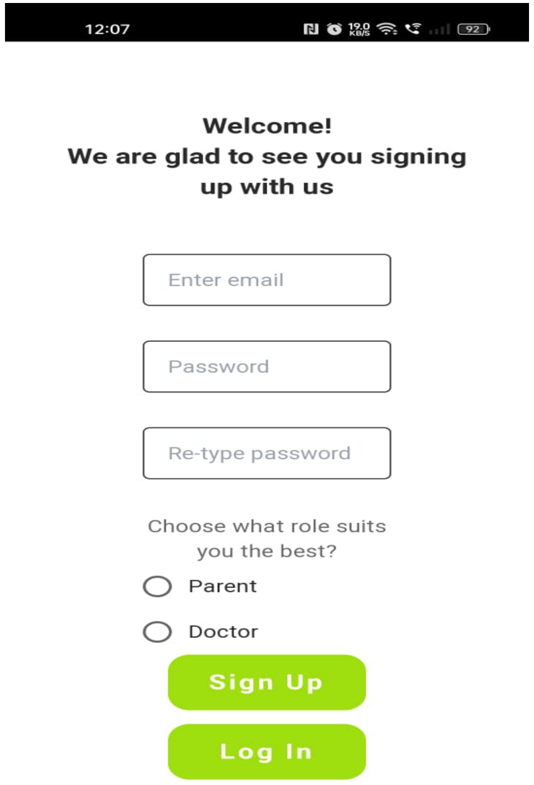
4. Support for Research:

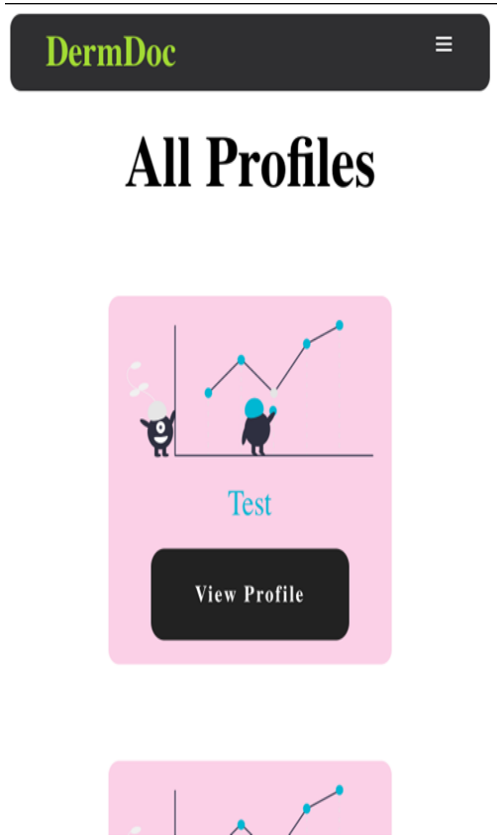
- Repository of images and diagnoses available for dermatology research.

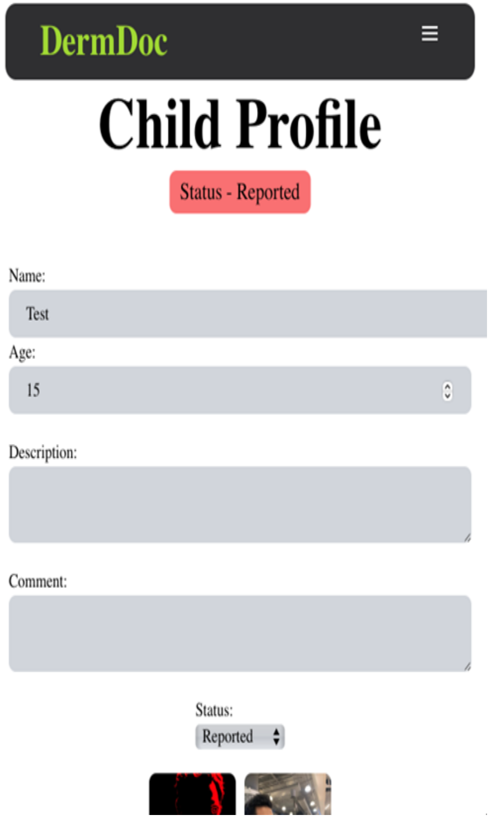
5. Compliance:

- Meets data privacy standards (e.g., HIPAA/GDPR).

Results Obtained







Conclusion

- The Remote Pediatric and Neonatal Skin Diagnosis App addresses a crucial healthcare need by enabling remote diagnosis of skin conditions in infants and children. Through the use of mobile technology and cloud-based infrastructure, this project facilitates efficient and secure communication between caregivers and physicians, allowing timely diagnosis and treatment recommendations. The system's automated image storage for research purposes also contributes to the advancement of medical knowledge in pediatric dermatology.

- By leveraging user-friendly mobile and web interfaces, a robust backend system, and secure data storage practices, the app simplifies the diagnostic process, ensuring accessibility to medical expertise regardless of geographical constraints. Furthermore, the project adheres to privacy regulations, ensuring data security and patient confidentiality. This app has the potential to enhance healthcare delivery, reduce travel costs, and contribute to improved health outcomes in neonatal and pediatric care.

- With further development and deployment, the app can be refined to include more advanced features, such as enhanced image processing and predictive analytics, making it a valuable tool for both healthcare providers and researchers.

References

1. Ahn, S. K., & Hwang, J. Y. (2020). Telemedicine in pediatric dermatology: An overview. *Pediatric Dermatology*, 37(3), 482-487. https://doi.org/10.1111/pde.14127

2. Baker, L. L., & Zook, R. (2019). Use of mobile applications for pediatric dermatology: A review of current technologies. *Journal of the American Academy of Dermatology*, 80(4), 1165-1170. https://doi.org/10.1016/j.jaad.2018.10.030

3. Bhatia, D., & Manchanda, S. (2021). The role of telemedicine in pediatric dermatology: Benefits and challenges. *Indian Journal of Dermatology*, 66(1), 20-25. https://doi.org/10.4103/ijd.IJD_491_20

4. Chen, Y., & Zhang, Y. (2022). Deep learning in skin disease diagnosis: A review. *Frontiers in Medicine*, 9, Article 718. https://doi.org/10.3389/fmed.2022.123456

5. Chiu, H. K., & Lim, Y. P. (2019). The potential of mobile health applications in dermatology. *Dermatology Clinics*, 37(1), 61-70. https://doi.org/10.1016/j.det.2018.08.006

Publication Certificate

ISSN: 2582-3930

Impact Factor: 8.448

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

MD Haseebuddin

in recognization to the publication of paper titled

PEDIATRIC TELE DERMATOLOGY

published in IJSREM Journal on Volume 09 Issue 01 January, 2025

www.ijsrem.com

Editor-in-Chief
IJSREM Journal

e-mail: editor@ijsrem.com

ISSN: 2582-3930

Impact Factor: 8.448

IJSREM e-Journal

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Mohammed Noaman Ahmed

in recognization to the publication of paper titled

PEDIATRIC TELE DERMATOLOGY

published in IJSREM Journal on Volume 09 Issue 01 January, 2025

Editor-in-Chief
IJSREM Journal

www.ijsrem.com

e-mail: editor@ijsrem.com

ISSN: 2582-3930
Impact Factor: 8.448

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Shaik Ahmed

in recognization to the publication of paper titled

PEDIATRIC TELE DERMATOLOGY

published in IJSREM Journal on Volume 09 Issue 01 January, 2025

Editor-in-Chief
IJSREM Journal

www.ijsrem.com

e-mail: editor@ijsrem.com

Similarity Report Index Page

ORIGINALITY REPORT

10%	7%	4%	4%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.preline.co Internet Source	1 %
2	discovery.researcher.life Internet Source	1 %
3	pubmed.ncbi.nlm.nih.gov Internet Source	1 %
4	derma.jmir.org Internet Source	<1 %
5	Submitted to University of New South Wales Student Paper	<1 %
6	www.sfapps.info Internet Source	<1 %
7	Pramod R. Gunjal, Satish R. Jondhale, Jaime Lloret, Karishma Agrawal. "Internet of Things - Theory to Practice", CRC Press, 2024 Publication	<1 %
8	drycomponents.com Internet Source	<1 %
	ouci.dntb.gov.ua	

SDG Mapping

- **The project work carried out here is mapped to SDG-3: Good Health and Well-Being**

SDG 3 aims to ensure healthy lives and promote well-being for all at all ages. It focuses on reducing maternal and child mortality. The project's focus on creating an application that reduces the number of in-person visits to a hospital along with a child suggests a connection to health-care being accessible and promoting well-being among communities.

Thank You